

CV 5101

Modeling, Uncertainty, and Data for Engineers

(July – Nov 2026)

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L7
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Flow

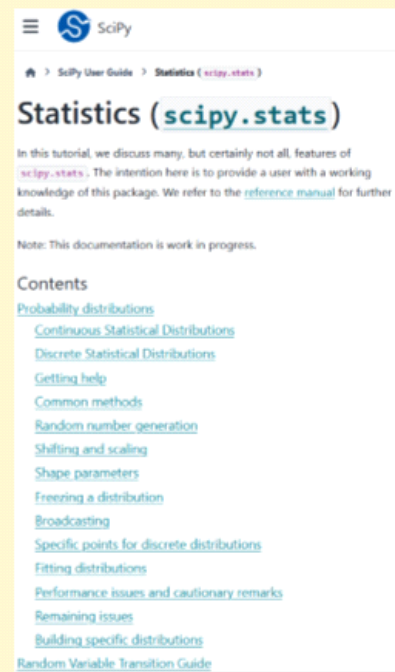
- Revise
 - Variance, covariance, **correlation**
 - Gaussian
 - reading z-table
 - Expectation
- Parametric distributions
 - Gaussian, Uniform, exponential, lognormal, Gumbel
- Propagation laws

Requested practice?

- Practice `scipy.stats` for the following:
 - Using distributions → Shifting, scaling
 - Generating random numbers
 - Fitting distributions

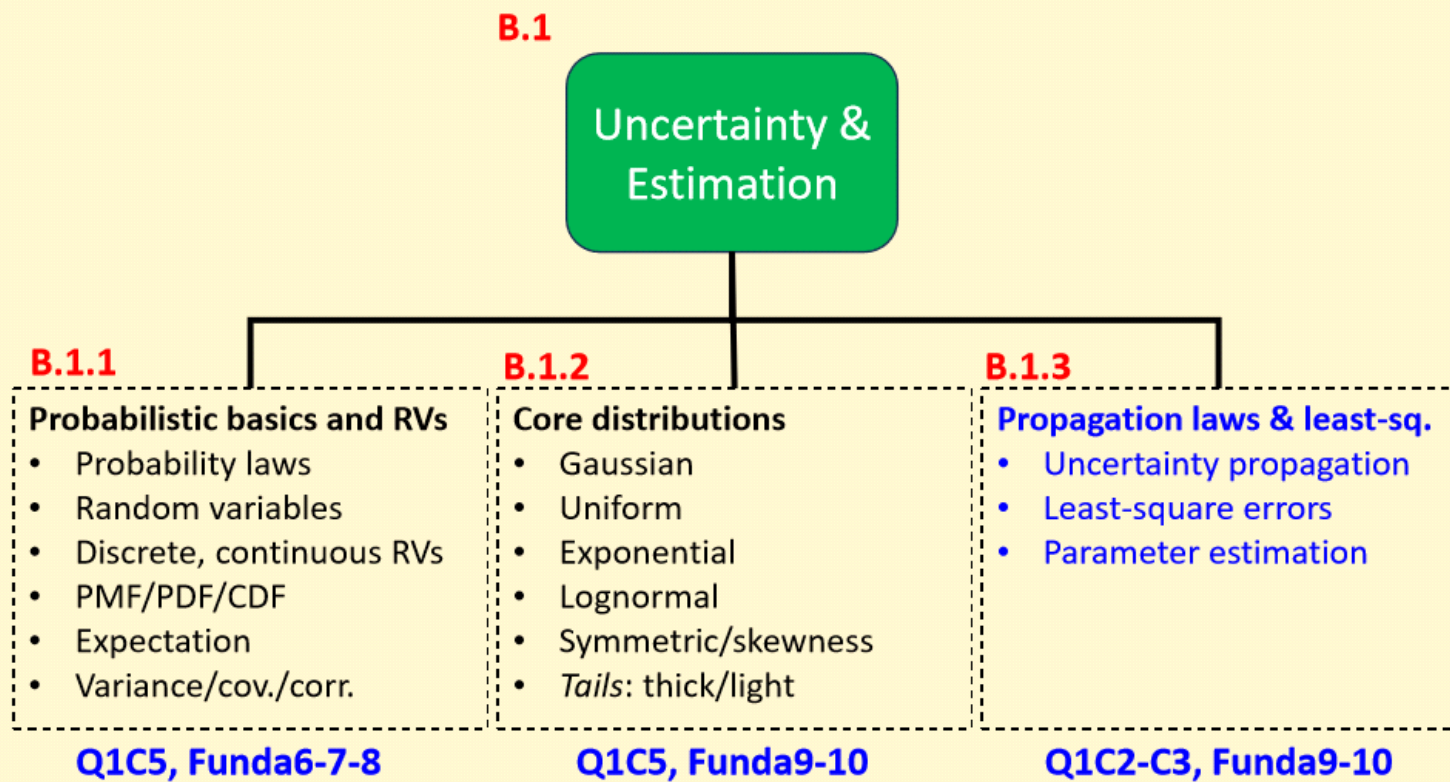


The screenshot shows the SciPy documentation page for Statistical functions (`scipy.stats`). The page title is "Statistical functions (`scipy.stats`)". Below the title, it states: "This module contains a large number of probability distributions, summary and frequency statistical tests, masked statistics, kernel density estimation, quasi-Monte Carlo functionality." It then lists the most important topics: `statsmodels` (regression, linear models, time series analysis), `Pandas` (tabular data, time series functionality), `PyMC` (Bayesian statistical modeling), `scikit-learn` (classification, regression, model selection), `Seaborn` (statistical data visualization), and `ipy` (Python to R bridge). Below this, there is a section for "Probability distributions" which explains that each univariate distribution is an instance of a subclass of `rv_continuous` or `rv_discrete`. It provides a table with three rows: `rv_continuous` (A generic continuous random variable), `rv_discrete` (A generic discrete random variable), and `rv_histogram` (Generates a distribution given by a histogram). Finally, there is a section for "Continuous distributions" with an example of `alpha` (An alpha continuous random variable).



The screenshot shows the SciPy documentation page for Statistics (`scipy.stats`). The page title is "Statistics (`scipy.stats`)". Below the title, it states: "In this tutorial, we discuss many, but certainly not all, features of `scipy.stats`. The intention here is to provide a user with a working knowledge of this package. We refer to the [reference manual](#) for further details." It then includes a note: "Note: This documentation is work in progress." Below this, there is a "Contents" section with a list of links: [Probability distributions](#), [Continuous Statistical Distributions](#), [Discrete Statistical Distributions](#), [Getting help](#), [Common methods](#), [Random number generation](#), [Shifting and scaling](#), [Shape parameters](#), [Freezing a distribution](#), [Broadcasting](#), [Specific points for discrete distributions](#), [Fitting distributions](#), [Performance issues and cautionary remarks](#), [Remaining issues](#), [Building specific distributions](#), and [Random Variable Transition Guide](#).

Module Overview



Revise

- Two structural engineers, Alice and Bob, independently measure the wind speed in m/s at the top of each tower.

Alice's reading $X \sim \mathcal{N}(\mu = 40, \sigma = 5)$

Bob's reading $Y \sim \mathcal{N}(\mu = 42, \sigma = 6)$

Their readings are correlated with $\rho = 0.8$

- If Alice and Bob average their measurements, $W = (X + Y)/2$, what are $E[W]$ and $\text{Var}[W]$?
- What's the covariance of X and Y ?
- If design wind speed is 55 m/s, what's the prob. that the average exceeds 55 m/s?

$$\begin{aligned} \mu_W &= E[W] = E\left[\frac{X+Y}{2}\right] \\ &= \frac{1}{2} E[X] + \frac{1}{2} E[Y] = 41 \end{aligned}$$



Revise

Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - **Q1C5 Univariate continuous distribution**
 - **Q1C5.1** PDF/CDF
 - **Q1C5.2** Empirical Distributions
 - **Q1C5.3** Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - Q1C6 Multivariate Distributions (briefly)
 - **Q1C2 Propagation of Uncertainty**
 - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
 - Q2C7 Extreme value theory: GEV, return period, POT
 - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
 - **Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables**
- Programming
 - Fundamental Concepts → Chapter 10

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B.1.3 Propagation laws & least-squares (Q1C2)

Uncertainty classification

Sources of Uncertainty

- **Inherent uncertainty**: concrete strength; irreducible
- **Statistical uncertainty**: due to the lack of data; reducible
- **Characteristics of system**: material properties, dimensions
- **Demand** placed on the system
- **Mathematical models** used to analyze structure's behavior
- **Measurements uncertainty**: while assessing structure's health
- **Probabilistic models** used to describe uncertain quantities
- **Human error**

Categories of Uncertainty

In classical statistics,

- Aleatory *Inherent, Irreducible*

epistemology

- **Epistemic**

*— Reducible
— Due to lack of knowledge*

Categories of Uncertainty

- Wind speed on the surface of a building:
 - Aleatory or epistemic?
- *Case 1: When there is no predictive wind model and only long-term statistical data is used?*
- *Case 2: When wind fields are studied using precise computational fluid dynamics?*

Propagation Laws

Mean and Variance propagation laws

- A random variable, $X \sim \mathcal{N}(40,5)$

If we define, another random variable, $Y = 3X + 10$, what are its mean and standard deviation (stdev) and type.

Recall: